

Status

HelixTranslate — Feature Status

Revision: 4 **Last modified:** 2026-06-15T20:10:00Z **Authority:** Derived from docs/features/.feature_inventory_raw.md (Revision 1). Per §11.4.45 (captured-evidence Status doc), §11.4.44 (revision header), §11.4.56 (two-audience summary → Status_Summary.md), §11.4.60 (always-sync), §11.4.6 (no-guessing — every status reflects what the inventory actually found in source). **Scope:** Every one of the 417 inventoried features across CLI, API, Web, gRPC, Library, Submodule and Infra. helix_qa deliberately not included.

Anti-bluff note (§11.4.6 / §11.4.2 / §11.4.107): Video-confirmation is set to a real file **only** where a recording exists AND that recording genuinely demonstrates the feature working (each cited .mp4 was ffprobe-verified for non-zero duration+frames and content-verified against its claimed behaviour — e.g. real DeepSeek output in the target language, real gRPC translated text, live WebSocket count change). **21 feature rows are video-confirmed** this session across unified-translator (DeepSeek translate, EPUB→TXT / HTML→EPUB / FB2→EPUB / FB2→FB2 conversions, Serbian-Cyrillic, -script Cyrillic[?]Latin, format-detection, help/version, -verify, -multipass multi-pass polishing engine), the REST API server (POST /api/v1/translate real DeepSeek + target_lang FIXED, TLS startup FIXED, /health · /api/v1/version · /api/v1/providers), the gRPC server (real EN→ES), markdown-translator (EPUB[?]MD round-trip), preparation-translator (FIXED real analysis), cmd/cli (real DeepSeek), monitor-server (live WS hub), and the DeepSeek library client. **Real-artifact-confirmed (not video): DOCX output** is now Implemented — a real DeepSeek run produced garden_es.docx = Microsoft Word 2007+ (well-formed WordprocessingML, real translation; pure-Go OOXML writer pkg/ebook/docx_writer.go, commit 87cd2be) — confirmed by the produced artifact's file type, no .mp4 exists yet (video PENDING). **Real-artifact / unit-confirmed (not video): PDF output** is now Implemented (commit fb265e7) — pkg/ebook/pdf_writer.go renders the Book→HTML5→weasyprint→valid Cyrillic-faithful PDF, wired into unified-translator (.pdf output; main.go:1160); pkg/ebook PDF tests green this session (no .mp4 yet, video PENDING). **Honest gaps recorded, NOT confirmed:** cmd/translator local path is a STUB (“local translation not yet implemented”); translate-ssh + ebook-translator are OPERATOR-BLOCKED (need a remote SSH/llama.cpp worker host); GET /api/v1/verified-models returns 404 because LLMsVerifier is disabled in config. Everything else is PENDING (a recording is owed) or N/A (no user-visible video applies — e.g. CLI flags, internal library types). Absence of a runtime recording is stated honestly, never papered over.

Operator-blocked items (read first, §11.4.45)

Two binaries are OPERATOR-BLOCKED — both require a remote SSH worker host running llama.cpp, which is not available in this environment. The binaries build and run as far as the missing host allows (captured in helixtranslate-cmd-blocked-binaries_20260615-172456.mp4):

- **cmd/translate-ssh** — SSH worker 4-step FB2→EPUB translator. **Unblock condition:** a reachable SSH host with llama.cpp + a model.
- **cmd/ebook-translator** — FB2 remote-translate workflow (syncs binary to remote, runs remote translate, downloads outputs). **Unblock condition:** same remote SSH/llama.cpp host.
- **cmd/translator (remote path)** — remote SSH/llama.cpp translate path is OPERATOR-BLOCKED (same host requirement); its LOCAL path is a separate STUB (“local translation not yet implemented”).

Coverage summary

Metric	Value
Headline feature total (inventory’s deduplicated per-category tally)	417
Enumerated feature rows in this doc	478 (470 source-inventory rows + 2 gap rows added Rev 2 [DOCX-write→Implemented Rev 3, PDF-write→Implemented Rev 4] + 6 new rows added Rev 4: cmd/model-bridge CLI ×2, pkg/bridge ×2, internal/verifier affirmative-gate ×1, pkg/api/dashboard.go ×1)
Implemented	471 (DOCX-write Rev 3 + PDF-write flipped gap→Implemented Rev 4 [commit fb265e7] + 6 new Rev 4 rows all Implemented)
Stub	4 (POST /api/v1/translate/ebook, POST /api/v1/preparation/analyze, GET /api/v1/preparation/result/:id, POST /api/v1/translate-with-verification)
Not implemented (gap rows)	0 (PDF-write flipped gap→Implemented Rev 4, commit fb265e7; output now EPUB/FB2/TXT/HTML/MD/DOCX/PDF)
Partial	3 (POST /api/batch on standalone pkg/api/server.go — returns queued batch_id only, no translation; vision_engine OpenCV — stub default, real impl behind build tag; cmd/translator — local path STUB + remote OPERATOR-BLOCKED)

Metric	Value
Operator-blocked	2 (translate-ssh, ebook-translator — need remote SSH/llama.cpp worker host)
Video-confirmed	21 (real, content-verified recordings this session — see Anti-bluff note) — $\approx$ 4.4% of 478 enumerated rows ($\approx$ 5.0% of the 417 headline). Plus 2 real-artifact/unit-confirmed (DOCX output — produced .docx = Microsoft Word 2007+; PDF output — pkg/ebook tests green + weasyprint present; both video PENDING). Rev-4 additions (bridge CLI+MCP, pkg/bridge, internal/verifier gate, pkg/api/dashboard.go) are PASS at unit/integration/live-run level with video PENDING (terminal-surface recordings owed) or SKIP (web dashboard — needs HelixQA web/video backend) or N/A (internal logic) — none claim a video that doesn't exist
Video-confirmation PENDING	runtime/user-visible features without a recording yet
Video-confirmation N/A	flags, internal library types, infra middleware (no standalone user-visible video)

§11.4.6 reconciliation: The inventory's headline is **417** — its own note states this is an *approximate, deduplicated* per-category tally where “some rows are sub-features... counted once, and a few rows span two categories (counted once, in the first/primary category).” The inventory's *detailed* tables actually enumerate **470** feature rows. This document transcribes **all 470 detailed rows 1:1** so **no feature is dropped** (which would itself be a bluff). The 417 headline and the 470 enumerated rows are both reported honestly; the 53-row gap is dedup, not missing coverage.

“Static-return” endpoints (e.g. GET /api/stats hardcoded zeros, GET /api/v1/status/:id hardcoded completed, GET /api/v1/metrics static, monitor-server GET /api/v1/status static monitoring_active) are marked **Implemented** at the route level but their Real-use cell states the static/hardcoded behaviour honestly so no false “working backend” impression is created.

Per-category counts

Category	Count
CLI	130
API	70
Web	23

Category	Count
gRPC	7
Library	81
Submodule	95
Infra	11
TOTAL	417

Legend

- **Implementation:** Implemented (real code path present) / Stub (returns mock, no real work) / Partial (does some but not the named work).
- **Wiring:** Wired (reachable from a binary / route / public API) / Unwired (present but not reached).
- **Real-use:** what the code actually does at runtime (honest, §11.4.6).
- **Tests:** Covered (unit/integration/challenge present in repo) / None (no test found / not asserted in inventory).
- **Validation:** strongest evidence available.
- **Video-confirmation:** Confirmed:<file> only with a real recording; else PENDING / N/A.

Per §11.4.6, where the inventory did not read test files line-by-line, Tests is recorded Not-inventoried (honest unknown) rather than guessed as Covered or None. The repo is known to carry extensive `_test.go` suites and a `test/` tree, but per-feature test attribution was not part of the raw inventory.

1. CLI Binaries — cmd/

cmd/api-server

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
api-server	REST translation server	API/Web	Implemented	Wired	Gin HTTP server (:8080) proxying to gRPC backend; serves dashboard + static files	Not-inventoried	S co (se m
api-server	-http-port flag	API	Implemented	Wired	Override HTTP port (else	Not-inventoried	S co

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Value
api-server	-http-address flag	API	Implemented	Wired	HTTP_PORT env) Override HTTP bind address (else HTTP_ADDRESS env)	Not-inventoried	Scalability
api-server	-grpc-address flag	API	Implemented	Wired	Override gRPC backend address (else GRPC_ADDRESS env)	Not-inventoried	Scalability
api-server	-debug flag	API	Implemented	Wired	Enable debug / Gin debug mode	Not-inventoried	Scalability

cmd/challenge-runner

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Value
challenge-runner	Challenge orchestrator	CLI	Implemented	Wired	Registers + runs all anti-bluff challenges, prints PASS/ FAIL/ SKIP, writes results dir	Not-inventoried	Scalability
challenge-runner	-parallel	CLI	Implemented	Wired	Run challenges in parallel	Not-inventoried	Scalability
challenge-runner	-max-concurrency	CLI	Implemented	Wired	Max concurrent challenges	Not-inventoried	Scalability
challenge-runner	-stop-on-failure	CLI	Implemented	Wired	Stop on first failure	Not-inventoried	Scalability
challenge-runner	-verbose	CLI	Implemented	Wired	Verbose output	Not-inventoried	Scalability
challenge-runner	-timeout	CLI	Implemented	Wired	Per-challenge timeout	Not-inventoried	Scalability
challenge-runner	-filter	CLI	Implemented	Wired	Comma-separated	Not-inventoried	Scalability

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Value
challenge-runner	-category	CLI	Implemented	Wired	challenge IDs to run Limit to a category	Not-inventoried	Score
challenge-runner	-results-dir	CLI	Implemented	Wired	Results output directory	Not-inventoried	Score

cmd/cli

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Value
cli	Universal ebook translator	CLI	Implemented	Wired	Parse ebook → translate → atomic-write EPUB/FB2/TXT; multi-llm/distributed support	Not-inventoried	Parse, run, Score, Time (d, sa, W, (8, va, w, si, os, su, pa, G, no, (5, pr
cli	-i/-input	CLI	Implemented	Wired	Input ebook (FB2/EPUB/TXT/HTML/PDF/DOCX)	Not-inventoried	Score
cli	-o/-output	CLI	Implemented	Wired	Output file	Not-inventoried	Score
cli	-f/-format	CLI	Implemented	Wired	Output format (epub, fb2, txt)	Not-inventoried	Score
cli	-p/-provider	CLI	Implemented	Wired	Translation provider	Not-inventoried	Score
cli	-model	CLI	Implemented	Wired		Not-inventoried	Score

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Verdict
					LLM model name		
cli	-api-key	CLI	Implemented	Wired	Provider API key	Not-inventoried	Score 100
cli	-base-url	CLI	Implemented	Wired	Provider base URL	Not-inventoried	Score 100
cli	-script	CLI	Implemented	Wired	Output script (default/cyrillic/latin)	Not-inventoried	Score 100
cli	-locale	CLI	Implemented	Wired	Target language locale code	Not-inventoried	Score 100
cli	-language	CLI	Implemented	Wired	Target language name	Not-inventoried	Score 100
cli	-source	CLI	Implemented	Wired	Source language (auto-detect if unset)	Not-inventoried	Score 100
cli	-detect	CLI	Implemented	Wired	Detect source language and exit (LLM detector if API key)	Not-inventoried	Score 100
cli	-v/-version	CLI	Implemented	Wired	Show version	Not-inventoried	Score 100
cli	-h/-help	CLI	Implemented	Wired	Show help	Not-inventoried	Score 100
cli	-create-config	CLI	Implemented	Wired	Write a config-file template	Not-inventoried	Score 100
cli	-disable-local-llms	CLI	Implemented	Wired	Use only distributed/API providers	Not-inventoried	Score 100
cli	-prefer-distributed	CLI	Implemented	Wired	Prefer distributed workers	Not-inventoried	Score 100
cli	-c/-config	CLI	Implemented	Wired	Config file path	Not-inventoried	Score 100
cli	-hash-codebase	CLI	Implemented	Wired	Compute codebase	Not-inventoried	Score 100

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
					hash and exit		

cmd/deployment

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
deployment	Distributed deployment manager	CLI	Implemented	Wired	Orchestrates Docker-based main+worker deployment over SSH	Not-invent
deployment	-config	CLI	Implemented	Wired	Config file (default config.distributed.json)	Not-invent
deployment	-action	CLI	Implemented	Wired	deploy/status/stop/cleanup/update/restart/generate-plan	Not-invent
deployment	-service	CLI	Implemented	Wired	Service name for update/restart	Not-invent
deployment	-image	CLI	Implemented	Wired	New image for update	Not-invent
deployment	-plan	CLI	Implemented	Wired	Deployment plan JSON file	Not-invent
deployment	-verbose	CLI	Implemented	Wired	Verbose logging	Not-invent
deployment	deploy/status/stop/cleanup/update/restart actions	CLI	Implemented	Wired	Execute deployment lifecycle ops	Not-invent
deployment	generate-plan action	CLI	Implemented	Wired	Generate deployment-plan.json (random JWT secret per instance)	Not-invent

cmd/ebook-translator

Component	Feature	Category	Implementation	Wiring	Real-use
ebook-translator	FB2 remote-translate workflow	CLI	Implemented	Wired	SSH worker syncs binary, runs remote FB2→MD→translate→EPUB, downloads + verifies outputs

Component	Feature	Category	Implementation	Wiring	Real-use
ebook-translator	Positional args (no flags)	CLI	Implemented	Wired	<source_fb2> <target_lang> <remote_host> <remote_user> <remote_password>
ebook-translator	Output verification	CLI	Implemented	Wired	Verifies MD/EPUB exist, non-degenerate, target-language (Cyrillic) check

cmd/grpc-server

Component	Feature	Category	Implementation	Wiring	Real-use	T
grpc-server	gRPC TranslationService	gRPC	Implemented	Wired	gRPC server (:50051): Start/Status/List/Cancel/Stream/GetProviders/SubscribeEvents	M i
grpc-server	-address	gRPC	Implemented	Wired	Bind address (or GRPC_ADDRESS env)	M i
grpc-server	-port	gRPC	Implemented	Wired	Port (or GRPC_PORT env)	M i
grpc-server	-max-connections	gRPC	Implemented	Wired	Max concurrent translations	M i
grpc-server	-reflection	gRPC	Implemented	Wired	Enable gRPC reflection (or ENABLE_REFLECTION env)	M i
grpc-server	-metrics	gRPC	Implemented	Wired	Enable metrics (or ENABLE_METRICS env)	M i
grpc-server	-log-level	gRPC	Implemented	Wired	Log level (or LOG_LEVEL env)	M i
grpc-server	-version	gRPC	Implemented	Wired	Show version	M i
grpc-server	-help	gRPC	Implemented	Wired	Show help	M i

cmd/markdown-translator

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
markdown-translator	EPUB [?] Markdown translate pipeline	CLI	Implemented	Wired	EPUB/MD → markdown → translate → EPUB/ MD; optional preparation phase	Not-inventoried
markdown-translator	-input	CLI	Implemented	Wired	Input file (EPUB or Markdown)	Not-inventoried
markdown-translator	-output	CLI	Implemented	Wired	Output file (auto if empty)	Not-inventoried
markdown-translator	-format	CLI	Implemented	Wired	Output format (epub, md)	Not-inventoried
markdown-translator	-lang	CLI	Implemented	Wired	Target language code	Not-inventoried
markdown-translator	-provider	CLI	Implemented	Wired	LLM provider (deepseek/ openai/ anthropic/ llamacpp)	Not-inventoried
markdown-translator	-model	CLI	Implemented	Wired	LLM model	Not-inventoried
markdown-translator	-keep-md	CLI	Implemented	Wired	Keep intermediate markdown	Not-inventoried
markdown-translator	-prepare	CLI	Implemented	Wired	Enable multi-LLM preparation/ analysis phase	Not-inventoried
markdown-translator	-prep-passes	CLI	Implemented	Wired	Number of preparation passes	Not-inventoried

cmd/monitor-server

Component	Feature	Category	Implementation	Wiring	Real-use
monitor-server	WebSocket monitoring hub	Web	Implemented	Wired	Gin server (:8090, MONITOR_SERVER_WS, /ws, /monitor dashboard status + health)
monitor-server	GET /ws	Web	Implemented	Wired	WebSocket subscription by session_id
monitor-server	/monitor + /	Web	Implemented	Wired	Serve monitor.html (redirect from /)
monitor-server	GET /api/v1/status/:session_id	Web/API	Implemented	Wired	Returns static monitoring_action JSON (not a live session lookup)
monitor-server	GET /health	Web	Implemented	Wired	Health + client count
monitor-server	MONITOR_SERVER_PORT env	Web	Implemented	Wired	Configure bind port (CLI flag)

cmd/preparation-translator

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
preparation-translator	Preparation+translation runner	CLI	Implemented	Wired	Parse ebook → multi-LLM preparation analysis → translate → write EPUB + analysis JSON	Not-invented
preparation-translator	-input	CLI	Implemented	Wired	Input ebook path	Not-invented
preparation-translator	-output	CLI	Implemented	Wired	Output EPUB path	Not-invented
preparation-translator	-analysis	CLI	Implemented	Wired	Preparation analysis JSON output path	Not-invented

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
preparation-translator	-source	CLI	Implemented	Wired	Source language	Not-invent
preparation-translator	-target	CLI	Implemented	Wired	Target language	Not-invent
preparation-translator	-passes	CLI	Implemented	Wired	Number of preparation passes	Not-invent
preparation-translator	-providers	CLI	Implemented	Wired	Comma-separated LLM providers	Not-invent

cmd/server

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
server	Main TLS HTTP/2+HTTP/3 server	API/Web	Implemented	Wired	TLS REST/WebSocket API (config.json); CORS + rate limit + JWT; optional distributed + LLMsVerifier	Not-inventoried
server	-config	API	Implemented	Wired	Config file path (default config.json)	Not-inventoried
server	-version	API	Implemented	Wired	Show version	Not-inventoried
server	-generate-certs	CLI	Implemented	Wired	Generate self-signed TLS certs (0600 key) and exit	Not-inventoried
server	HTTP/3 (QUIC) + HTTP/2 fallback serving	API/Web	Implemented	Wired	Serve API over TLS; HTTP/3 if EnableHTTP3	Not-inventoried
server	API routes (pkg/api Handler)	API/Web	Implemented	Wired	Registers translation REST +	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
server	LLMsVerifier routes (if enabled)	API	Implemented	Wired	WebSocket routes Registers verifier endpoints	Not-inventoried

cmd/ssh-translation

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
ssh-translation	SSH translation system (fixed-config)	CLI/Web	Implemented	Wired	Hardcoded Russian→Serbian-Cyrillic workflow over SSH worker w/ llama.cpp; WebSocket monitoring	Not-inventoried
ssh-translation	(no CLI flags)	CLI	Implemented	Wired	All config hardcoded in SystemConfig (host, creds, langs, llama providers)	Not-inventoried

cmd/translate-ssh

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
translate-ssh	SSH worker 4-step FB2→EPUB translator	CLI	Implemented	Wired	FB2→MD → translate (remote llama.cpp) → MD→EPUB; standalone worker invoked by coordinator	Not-inventoried
translate-ssh	-input	CLI	Implemented	Wired	Input ebook (required)	Not-inventoried
translate-ssh	-output	CLI	Implemented	Wired	Output EPUB (required)	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
translate-ssh	-host	CLI	Implemented	Wired	SSH host (required)	Not-inventoried	S
translate-ssh	-user	CLI	Implemented	Wired	SSH username (required)	Not-inventoried	S
translate-ssh	-password	CLI	Implemented	Wired	SSH password (required)	Not-inventoried	S
translate-ssh	-port	CLI	Implemented	Wired	SSH port	Not-inventoried	S
translate-ssh	-remote-dir	CLI	Implemented	Wired	Remote working directory	Not-inventoried	S
translate-ssh	-report-dir	CLI	Implemented	Wired	Report output directory	Not-inventoried	S

cmd/translator

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
translator	Documented FB2 local/remote translator	CLI	Partial (local path is STUB; remote needs SSH worker)	Wired	Remote SSH/llama.cpp path produces a documentation report; LOCAL path is a STUB — emits “local translation not yet implemented” (captured). Remote path is OPERATOR-BLOCKED (no SSH worker host)	Not-inventoried	S
translator	-i/-input	CLI	Implemented	Wired	Input ebook	Not-inventoried	S
translator	-o/-output	CLI	Implemented	Wired	Output file	Not-inventoried	S
translator	-ssh-host	CLI	Implemented	Wired	SSH host (remote mode)	Not-inventoried	S
translator	-ssh-user	CLI	Implemented	Wired	SSH username	Not-inventoried	S
translator		CLI	Implemented	Wired	SSH password		S

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
	-ssh-password					Not-inventoried
translator	-ssh-port	CLI	Implemented	Wired	SSH port	Not-inventoried
translator	-remote-dir	CLI	Implemented	Wired	Remote working directory	Not-inventoried
translator	-workers	CLI	Implemented	Wired	Parallel workers	Not-inventoried
translator	-chunk-size	CLI	Implemented	Wired	Text chunk size	Not-inventoried
translator	-concurrency	CLI	Implemented	Wired	Max concurrent operations	Not-inventoried
translator	-verify	CLI	Implemented	Wired	Verify translated output	Not-inventoried
translator	-verbose	CLI	Implemented	Wired	Verbose logging	Not-inventoried
translator	-llama-binary	CLI	Implemented	Wired	Path to llama.cpp binary	Not-inventoried
translator	-temperature	CLI	Implemented	Wired	LLM temperature	Not-inventoried
translator	-context	CLI	Implemented	Wired	LLM context size	Not-inventoried
translator	-version	CLI	Implemented	Wired	Show version	Not-inventoried
translator	-help	CLI	Implemented	Wired	Show help	Not-inventoried
translator	-hash-codebase	CLI	Implemented	Wired	Compute codebase hash	Not-inventoried

cmd/unified-translator (PRIMARY CLI)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
unified-translator	Primary multi-provider translator	CLI	Implemented	Wired	Parse → MD → translate (API/llamacpp/ssh) → script-normalize → output (epub/fb2/html/txt/md) + session report	Not-inven

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
unified-translator	DeepSeek translate (provider=deepseek)	CLI	Implemented	Wired	EN→Serbian DeepSeek run, real LLM output written to TXT	Not-invented
unified-translator	EPUB→TXT conversion	CLI	Implemented	Wired	EPUB parse → DeepSeek translate → TXT writer, real Serbian text	Not-invented
unified-translator	HTML→EPUB conversion	CLI	Implemented	Wired	HTML parse → DeepSeek translate → EPUB writer	Not-invented
unified-translator	FB2→EPUB conversion	CLI	Implemented	Wired	FB2 parse → DeepSeek translate → EPUB writer	Not-invented
unified-translator	FB2→FB2 conversion	CLI	Implemented	Wired	FB2 parse → DeepSeek translate → FB2 writer (namespaces preserved)	Not-invented
unified-translator	Serbian Cyrillic output	CLI	Implemented	Wired	Translate with Serbian-Cyrillic target output	Not-invented
unified-translator	Format detection	CLI	Implemented	Wired	Input-format auto-detection on real input	Not-invented
unified-translator	help/version output	CLI	Implemented	Wired	–help/–version usage banner	Not-invented
unified-translator	–i/–input	CLI	Implemented	Wired	Input ebook	Not-invented
unified-translator	–o/–output	CLI	Implemented	Wired	Output file (auto if empty)	Not-invented
unified-translator	–source–lang	CLI	Implemented	Wired	Source language	Not-invented
unified-translator	–target–lang	CLI	Implemented	Wired	Target language	Not-invented
unified-translator	–script (Serbian Cyrillic↔Latin)	CLI	Implemented	Wired	Target script (cyrillic/latin) script	Not-invented

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					normalization on real run	
unified-translator	-provider	CLI	Implemented	Wired	Provider (openai/anthropic/hipu/deepseek/qwen/gemini/ollama/llamacpp/ssh)	Not-invented
unified-translator	-model	CLI	Implemented	Wired	Model name	Not-invented
unified-translator	-api-key	CLI	Implemented	Wired	Provider API key	Not-invented
unified-translator	-base-url	CLI	Implemented	Wired	Provider base URL	Not-invented
unified-translator	-temperature	CLI	Implemented	Wired	LLM temperature	Not-invented
unified-translator	-max-tokens	CLI	Implemented	Wired	Max tokens	Not-invented
unified-translator	-timeout	CLI	Implemented	Wired	Request timeout	Not-invented
unified-translator	-ssh-host	CLI	Implemented	Wired	SSH host (provider=ssh)	Not-invented
unified-translator	-ssh-user	CLI	Implemented	Wired	SSH username	Not-invented
unified-translator	-ssh-password	CLI	Implemented	Wired	SSH password	Not-invented
unified-translator	-ssh-port	CLI	Implemented	Wired	SSH port	Not-invented
unified-translator	-remote-dir	CLI	Implemented	Wired	Remote directory	Not-invented
unified-translator	-llama-binary	CLI	Implemented	Wired	Path to llama.cpp binary	Not-invented
unified-translator	-llama-model	CLI	Implemented	Wired	Path to llama.cpp model	Not-invented
unified-translator	-context-size	CLI	Implemented	Wired	LLM context size	Not-invented
unified-translator	-workers	CLI	Implemented	Wired	Parallel workers	Not-invented
unified-translator	-chunk-size	CLI	Implemented	Wired	Text chunk size	Not-invented
unified-translator	-concurrency	CLI	Implemented	Wired	Concurrent operations	Not-invented

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
unified-translator	-verify (output-quality check)	CLI	Implemented	Wired	Built-in per-step verification: input/markdown/translation-quality/valid-EPUB verdicts in session report. NOTE: this is the unified-translator's own verify pass, NOT the pkg/verification multipass engine (that engine is not wired into this CLI — see gap row below)	Not-inven
unified-translator	-verbose	CLI	Implemented	Wired	Verbose logging	Not-inven
unified-translator	-monitoring	CLI/Web	Implemented	Wired	Enable web monitoring	Not-inven
unified-translator	-monitoring-port	CLI/Web	Implemented	Wired	Monitoring server port	Not-inven
unified-translator	-use-verifier	CLI	Implemented	Wired	Use LLMsVerifier for model selection	Not-inven
unified-translator	-verifier-url	CLI	Implemented	Wired	LLMsVerifier API URL	Not-inven
unified-translator	-verifier-api-key	CLI	Implemented	Wired	LLMsVerifier API key	Not-inven
unified-translator	-version	CLI	Implemented	Wired	Show version	Not-inven
unified-translator	-help	CLI	Implemented	Wired	Show help	Not-inven

cmd/verify-models

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
		CLI	Implemented	Wired	Runs verification		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Ver
verify-models	In-process LLMsVerifier runner				pipeline against env-keyed providers; persists verified models to SQLite	Not-inventoried	Sc
verify-models	-db	CLI	Implemented	Wired	SQLite store path	Not-inventoried	Sc
verify-models	-max	CLI	Implemented	Wired	Max models/provider through pipeline	Not-inventoried	Sc
verify-models	-min-score	CLI	Implemented	Wired	Min overall score to persist	Not-inventoried	Sc
verify-models	-timeout	CLI	Implemented	Wired	Overall verification timeout	Not-inventoried	Sc

cmd/model-bridge

LLMsVerifier→component+agent bridge (no local models): selects the best LLMsVerifier-scored model, exposes it to agents/components via a CLI and an MCP stdio server wired in `.mcp.json`. Source: `cmd/model-bridge/{main.go,mcp.go}`, `pkg/bridge/`, commit `ab1bed3` (+ routing seam `a5860b2`).

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
model-bridge	LLMsVerifier→agent bridge CLI	CLI	Implemented	Wired	Selects best LLMsVerifier-scored model (top-1 + fallback chain) and translates/invokes through it; no local models	Coverage: bridge unit+go t greener sessi
model-bridge	MCP stdio server (<code>.mcp.json</code>)	CLI/Infra	Implemented	Wired	Exposes the selected model to MCP clients over stdio; wired as <code>model-</code>	Coverage: model mcp_

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					bridge in .mcp.json (./build/ model- bridge)	

cmd/workable-items

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
workable-items	§11.4.93 workable-items SQLite tool	CLI	Implemented	Wired	Bidirectional sync between Issues/Fixed markdown and SQLite single- source-of-truth	Not- inventoried
workable-items	sync md- to-db	CLI	Implemented	Wired	Parse ATM-NNN headings from markdown into DB	Not- inventoried
workable-items	sync db- to-md	CLI	Implemented	Wired	Regenerate Issues_Summary/ Fixed_Summary from DB	Not- inventoried
workable-items	validate	CLI	Implemented	Wired	Verify DB matches markdown; exit non-zero on drift	Not- inventoried
workable-items	diff	CLI	Implemented	Wired	Print md-vs-db drift human- readably	Not- inventoried
workable-items	list	CLI	Implemented	Wired	Print items in DB	Not- inventoried
workable-items	record- event	CLI	Implemented	Wired	Append event to item_history audit trail	Not- inventoried
workable-items	path/ value flags	CLI	Implemented	Wired	-issues/-fixed/-db/-issues- summary/-fixed- summary/-atm/-event/-on/-not	Not- inventoried

2. Library Packages — pkg/

pkg/translator/llm — LLM provider clients & factory

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
LLM Core	LLMClient interface & factory	Library	Implemented	Wired	LLMClient interface + NewLLMTranslator factory dispatching to ~32 providers	Not-invented
LLM Core	Type-coercion utilities	Library	Implemented	Wired	JSON config conversion helpers	Not-invented
LLM Core	Verified factory	Library	Implemented	Wired	LLMsVerifier-gated translator construction w/ fallback chain (CONST-034)	Not-invented
LLM Core	Mock client	Library	Implemented	Wired	Test mock LLMClient with fault injection (test-only)	Not-invented
OpenAI	OpenAI client	Library	Implemented	Wired	GPT chat-completions translation	Not-invented
Anthropic	Anthropic Claude client	Library	Implemented	Wired	Claude /v1/ messages translation	Not-invented
Google	Gemini client	Library	Implemented	Wired	Google Gemini generativelanguage API	Not-invented
DeepSeek	DeepSeek client	Library	Implemented	Wired	DeepSeek OpenAI-compatible chat-completions	Not-invented
Zhipu	Zhipu (GLM) client	Library	Implemented	Wired	Zhipu GLM /api/paas/v4	Not-invented
Alibaba	Qwen client	Library	Implemented	Wired	Qwen/DashScope (OAuth2-capable)	Not-invented
Ollama	Ollama local client	Library	Implemented	Wired	Local /api/generate inference	Not-invented
llama.cpp	llama.cpp client	Library	Implemented	Wired	llama-cli wrapper w/ hardware	Not-invented

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					detection, model download, GPU accel	
llama.cpp	llama.cpp multi-worker coordinator	Library	Implemented	Wired	Worker pool for parallel local llama.cpp inference	Not-invented
Groq	Groq client	Library	Implemented	Wired	OpenAI-compatible wrapper	Not-invented
Cohere	Cohere client	Library	Implemented	Wired	command-r via OpenAI-compatible endpoint	Not-invented
Mistral	Mistral client	Library	Implemented	Wired	OpenAI-compatible wrapper	Not-invented
xAI	xAI (Grok) client	Library	Implemented	Wired	OpenAI-compatible wrapper	Not-invented
Replicate	Replicate client	Library	Implemented	Wired	Async prediction-polling API	Not-invented
Cerebras	Cerebras client	Library	Implemented	Wired	OpenAI-compatible wrapper (base-URL/model specifics UNCONFIRMED)	Not-invented
Cloudflare	Cloudflare Workers AI client	Library	Implemented	Wired	/client/v4 Workers AI	Not-invented
SiliconFlow	SiliconFlow client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-invented
Hyperbolic	Hyperbolic client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-invented
TogetherAI	TogetherAI client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-invented
SambaNova	SambaNova client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-invented
Kimi	Kimi (Moonshot) client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-invented
Novita	Novita client	Library	Implemented	Wired	/v3/openai-compatible wrapper	Not-invented
NLPCloud	NLPCloud client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-invented
Upstage		Library	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
	Upstage (Solar) client				OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-inventor
Sarvam	Sarvam client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-inventor
Modal	Modal client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-inventor
PublicAI	PublicAI client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-inventor
NIA	NIA client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-inventor
Vulavula	Vulavula client	Library	Implemented	Wired	OpenAI-compatible wrapper (specifics UNCONFIRMED)	Not-inventor

pkg/translator — translation core

Component	Feature	Category	Implementation	Wiring	Real-use
Translator Core	Translator interface + BaseTranslator	Library	Implemented	Wired	Translate/ TranslateWithProgress GetStats + base impl v caching & stats
Translator Core	Translation config	Library	Implemented	Wired	TranslationConfig (provider/model/key/ter tokens/lang-pair/script)
Translator Core	UniversalTranslator	Library	Implemented	Wired	High-level ebook translator: lang detection per-chapter translation, progress events

pkg/ebook — format parsers & writers

Component	Feature	Category	Implementation	Wiring	Real-use
ebook	Universal parser dispatcher	Library	Implemented	Wired	Multi-format parser (FB2/EPUB/...
ebook	FB2 parse	Library	Implemented	Wired	Parse FictionBook 2.0 → Book, r
ebook	FB2 write	Library	Implemented	Wired	Write Book → FB2 XML

Component	Feature	Category	Implementation	Wiring	Real-use
ebook	EPUB parse	Library	Implemented	Wired	Parse EPUB (ZIP+OPF/NCX)
ebook	EPUB write	Library	Implemented	Wired	Write Book → EPUB 2.0 ZIP
ebook	DOCX parse	Library	Implemented	Wired	Parse .docx (archive/zip + encoding/xml)
ebook	HTML parse	Library	Implemented	Wired	Parse HTML → Book
ebook	TXT parse	Library	Implemented	Wired	Parse plain text (configurable 64K)
ebook	PDF text extraction	Library	Implemented	Wired	Extract text via ledongthuc/pdf
ebook	DOCX write (output)	Library	Implemented	Wired	Pure-Go OOXML writer pkg/ebook/docx_wri archive/zip + encoding/xml, zero OOXML parts; XML specials auto-chardata. Wired into unified-translator (docx). Real DeepSeek EN→ES – Microsoft Word 2007+, well-fitted the real translation. Guards: pkg/ebook/docx_wri_test (cmd/unified-translator/proof-proven)
ebook	PDF write (output)	Library	Implemented	Wired	PDF writer pkg/ebook/pdf_wri renders the in-memory Book to UML. WeasyPrint to emit a VALID, CJK fonts can't render Cyrillic — weasyprint silent-glyph-drop bluff). Wired into generateOutput (case .pdf → epub, main.go:1160); .pdf is a supported type ErrWeasyPrintUnavailable (§11.4.3/§11.4.6). Guards: pkg/ebook/pdf_wri_test (TestPDFWriter_ProducesValid_UnavailableWeasyPrint)

pkg/fb2 — FB2-specific XML

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
fb2	FB2 parser (mixed-content)	Library	Implemented	Wired	Parse FB2 preserving paragraphs/verses/inline mixed content	Not-inventoried
fb2	FB2→Markdown converter	Library	Implemented	Wired	Convert FB2 → Markdown (author/	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					annotation/ epigraph/ poem/cite)	

pkg/format

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
format	Format detector	Library	Implemented	Wired	Detect format by magic bytes/extension/content; ZIP disambiguation (EPUB/DOCX/AZW3)	Not-inventoried	Sou con

pkg/markdown — EPUB↔Markdown workflow

Component	Feature	Category	Implementation	Wiring	Real-use
markdown	EPUB→Markdown converter	Library	Implemented	Wired	EPUB→MD w/ YAML image extraction, GFM
markdown	Markdown→EPUB converter	Library	Implemented	Wired	MD→EPUB 2.0 w/ OP tables/blockquotes
markdown	Simple workflow	Library	Implemented	Wired	ebook→markdown→tra pipeline w/ progress
markdown	Markdown translator	Library	Implemented	Wired	Line-by-line MD transla preserving formatting/c

pkg/script

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
script	Cyrillic↔Latin conversion	Library	Implemented	Wired	Bidirectional Serbian script conversion, context-aware digraphs	Not-inventoried

pkg/storage

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valida
storage	Storage interface	Library	Implemented	Wired	Abstract session + translation-cache persistence	Not-inventoried	Source confirmed
storage	PostgreSQL backend	Library	Implemented	Wired	Pooled Postgres w/ UNIQUE-tuple dedup	Not-inventoried	Source confirmed
storage	Redis backend	Library	Implemented	Wired	Redis cache, sha256-hashed tuple keys (collision-proof)	Not-inventoried	Source confirmed
storage	SQLite backend	Library	Implemented	Wired	SQLite (optional SQLCipher) w/ UNIQUE-tuple migrations	Not-inventoried	Source confirmed
storage	Cache-key hashing	Library	Implemented	Wired	Length-prefixed tuple encoding for collision-resistant keys	Not-inventoried	Source confirmed

pkg/security

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valida
security	JWT auth service	Library	Implemented	Wired	JWT generate/validate/refresh w/ timing-attack mitigation	Not-inventoried	Source confirmed
security		Library	Implemented	Wired	Thread-safe key	Not-inventoried	Source confirmed

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valida
	API-key store				validation w/ expiry & revocation		
security	Rate limiter	Library	Implemented	Wired	Token-bucket per-key limiting w/ cleanup & stats	Not-inventoried	Source confirm
security	User auth service	Library	Implemented	Wired	Login/user-create w/ timing & enumeration mitigation	Not-inventoried	Source confirm

pkg/verification — quality verification & polishing

Component	Feature	Category	Implementation	Wiring	Real-use
verification	Heuristic verifier	Library	Implemented	Wired	Untranslated-block/HTML-artifact detection, quality sc
verification	Multi-pass polisher	Library	Implemented	Wired into unified-translator (-multipass, d53e085)	Iterative multi-LLM polishing, note-taking & consensus. N reachable via the unified-translator opt-in -multipass flag (+ -multipass-passe -multipass-db; default O zero overhead/behavior cha when absent). NOTE: distin from -verify (the CLI's ov per-step output check) — -multipass invokes this p verification engine. Gua cmd/unified-translator/multipass_wiring_test.go (httptest + real DeepSeek). §11.4.150 research: docs/research/20260615_video_surfaced_verification_multipass_wiri
verification	Book polisher	Library	Implemented	Wired	Multi-LLM per-dimension scoring & consensus valida
verification	Polishing reporter	Library	Implemented	Wired	Stats aggregation (consensu rates, issue/provider breakd
verification	Polishing database	Library	Implemented	Wired	SQLite persistence for mult sessions/pass records
verification		Library	Implemented	Wired	

Component	Feature	Category	Implementation	Wiring	Real-use
	Literary notes				Character/tone/theme/terminology/cultural note collection

internal/verifier — LLMsVerifier consumer integration

Consumer-side LLMsVerifier verification gate (client + scoring + discovery + selection). Source: `internal/verifier/pipeline.go`.

Component	Feature	Category	Implementation	Wiring	Real-use
internal/verifier	Affirmative-response hard-gate	Library	Implemented	Wired (verification pipeline)	Correctness fix (commit 97a8afd): a model with <code>affirmative_response=</code> is now a HARD disqualification — rejected by the verification gate regardless of other scores

pkg/preparation — pre-translation analysis

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
preparation	Preparation coordinator	Library	Implemented	Wired	Multi-pass content analysis (character/terminology/cultural)	Not-inventoried	S c
preparation	Preparation-aware translator	Library	Implemented	Wired	Wraps base translator w/ analysis phase + context passing	Not-inventoried	S c
preparation	Prompt builder	Library	Implemented	Wired	Context-aware analysis prompts	Not-inventoried	S c

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
preparation	Domain types	Library	Implemented	Wired	refined across passes Content-classification/ character/ cultural/ terminology structs	Not-inventoried	S c
preparation	Utilities	Library	Implemented	Wired	JSON serialization + human-readable summaries	Not-inventoried	S c

pkg/coordination — multi-LLM coordination

Component	Feature	Category	Implementation	Wiring	Real-use	Te
coordination	Multi-LLM coordinator	Library	Implemented	Wired	Multiple LLM instances w/ round-robin, retry, distributed fallback	No inv
coordination	Multi-LLM translator wrapper	Library	Implemented	Wired	Adapts coordinator to translator.Translator interface	No inv

pkg/distributed — SSH distributed workers

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
distributed	Coordinator (distribute + aggregate)	Library	Implemented	Wired	Distributes work across remote LLM instances, round-robin, failover to local	Not-inventoried
distributed	SSH connection pool	Library	Implemented	Wired	Pooled SSH connections w/ idle-	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
distributed	Worker pairing/discovery	Library	Implemented	Wired	timeout cleanup Discover/ pair/health-check remote translator services	Not-inventoried
distributed	Fallback / graceful degradation	Library	Implemented	Wired	Retry w/ backoff, degraded-mode entry/exit, recovery tracking	Not-inventoried
distributed	Version manager	Library	Implemented	Wired	Version check/ update/ rollback, signed packages, backups, drift alerts (no LLM)	Not-inventoried
distributed	Performance instrumentation	Library	Implemented	Wired	Conn pooling, TTL result cache, circuit breaker, batch processing	Not-inventoried
distributed	Security hardening	Library	Implemented	Wired	SSH host-key verify, TLS validation, network ACLs, security event log	Not-inventoried
distributed	Manager / orchestration	Library	Implemented	Wired	Init/ discover/ pair/ version-manage + translation delegation	Not-inventoried

pkg/sshworker

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valida
sshworker	SSH worker	Library	Implemented	Wired	SSH exec, file up/download, codebase sync, key generation	Not-inventoried	Source confir
sshworker	Monitored worker	Library	Implemented	Wired	Event-monitored progress tracking for long commands	Not-inventoried	Source confir

pkg/deployment

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
deployment	Deployment orchestrator	Library	Implemented	Wired	Automated multi-host instance deployment	Not-inver
deployment	Docker orchestrator	Library	Implemented	Wired	Generate/manage docker-compose for multi-service deploy	Not-inver
deployment	SSH deployer	Library	Implemented	Wired	SSH-based deploy w/ validation & error handling	Not-inver
deployment	Network discovery	Library	Implemented	Wired	UDP service discovery/broadcast	Not-inver
deployment	API comm logger	Library	Implemented	Wired	Retrofit-style REST request/response logging between nodes	Not-inver
deployment	Deployment types	Library	Implemented	Wired	DeploymentPlan/Status/NetworkService/APICommunicationLog	Not-inver

pkg/grpc

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
grpc	gRPC TranslationService	Library	Implemented	Wired	gRPC server: sessions, event	Not-inventorie

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
grpc	Core translator pipeline	Library	Implemented	Wired	streaming, provider registry, concurrency Translation job exec w/ step tracking + progress via llm factory	Not-inventoried

pkg/api (library layer)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
api	REST API server	Library	Implemented	Wired	HTTP server: auth, CSRF, rate-limit, routing	Not-inventoried	Sc
api	Translation endpoints	Library	Implemented	Wired	text/FB2/ ebook translate, batch ops, lang/provider listing	Not-inventoried	Sc
api	Batch translation handlers	Library	Implemented	Wired	Batch requests w/ queue, progress, concurrency	Not-inventoried	Sc
api	Verifier endpoints	Library	Implemented	Wired	LLMsVerifier model verification/ scoring/ selection; / verified-models	Not-inventoried	Sc

pkg/events, websocket, batch, progress, report, logger, hardware, hash

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
events	Event pub/sub bus	Library	Implemented	Wired	Thread-safe pub/sub w/	Not-inventoried	Sou

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
websocket	WebSocket hub	Library	Implemented	Wired	subscription IDs + type filtering Real-time session event broadcast to dashboards	Not-inventoried	Sou con
batch	Batch processor	Library	Implemented	Wired	File/dir translation w/ concurrency & progress	Not-inventoried	Sou con
progress	Progress tracker	Library	Implemented	Wired	Chapter/section progress w/ ETA	Not-inventoried	Sou con
report	Report generator	Library	Implemented	Wired	Session reporting (issues/warnings/logs/results)	Not-inventoried	Sou con
logger	Structured logger	Library	Implemented	Wired	Text/JSON logging w/ levels	Not-inventoried	Sou con
hardware	Hardware detector	Library	Implemented	Wired	CPU/RAM/GPU detection + model-size estimation	Not-inventoried	Sou con
hash	Codebase hash generator	Library	Implemented	Wired	SHA256 of source/config/deps/binaries	Not-inventoried	Sou con

pkg/language

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Validation
language	Language registry / heuristic detector	Library	Implemented	Wired	ISO 639-1 registry + heuristic detection	Not-inventoried	Source-confirmed

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Validation
language	LLM language detector	Library	Implemented	Wired	(18+ langs) LLM-driven language detection	Not-inventoried	Source-confirmed

pkg/models & modelsbridge

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
models	Model registry	Library	Implemented	Wired	Translation-model metadata registry (50+ models)	Not-inventoried
models	Model downloader	Library	Implemented	Wired	HTTP model download w/ caching + sha256 verification	Not-inventoried
models	User domain	Library	Implemented	Wired	User struct, bcrypt hashing, in-memory repo	Not-inventoried
models	Error definitions	Library	Implemented	Wired	Domain error constants	Not-inventoried
models	Verifier registry	Library	Implemented	Wired	LLMsVerifier-backed registry w/ thresholds & filtering	Not-inventoried
modelsbridge	Models bridge	Library	Implemented	Wired	Adapter mapping models.LLMRequest/Response → llm.LLMClient	Not-inventoried

pkg/bridge — LLMsVerifier→component+agent bridge

In-process bridge (no local models) that selects the best LLMsVerifier-scored model and routes component/agent calls to it; backs cmd/model-bridge CLI + MCP.
Source: pkg/bridge/bridge.go, commit ab1bed3 (+ routing mutation seam a5860b2).

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
bridge	Best-model selection	Library	Implemented	Wired (via cmd/model-bridge)	Picks top-1 LLMsVerifier-scored model with a	Covered (pkg/unit+integration)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
bridge	Invoke routes to BestModel	Library	Implemented	bridge CLI+MCP)	fallback chain; no local models	Covered + 8 TestBridge (RED FAIL
				Wired	Routes the invocation to the selected best model	

pkg/version

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valida
version	App version	Library	Implemented	Wired	Singleton version constant (2.3.0)	Not-inventoried	Source confirm
version	Codebase hasher	Library	Implemented	Wired	SHA256 codebase versioning over dirs/ extensions w/ exclude patterns	Not-inventoried	Source confirm

pkg/challenge_runner — anti-bluff test harness (uses digital.vasic.challenges)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
challenge_runner	Challenge orchestrator	Library	Implemented	Wired	Coordinates 18+ integration challenges, parallel exec + filtering	Not-inventori
challenge_runner	Build & format-detection challenges	Library	Implemented	Wired	Build-verification + FB2/EPUB/ TXT/HTML/ PDF/DOCX format-	Not-inventori

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
challenge_runner	Infrastructure challenges	Library	Implemented	Wired	detection challenges Event-bus + other infra challenges (run go test)	Not-inventoried
challenge_runner	Provider challenges	Library	Implemented	Wired	Verify LLM provider factory/interface compile+load (build, not live calls)	Not-inventoried

3. Web Dashboards & Endpoints

Web Dashboards

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
monitor.html	WebSocket connection status	Web	Implemented	Wired	Green/red pulse dot + Connected/Disconnected indicator	Not-inventoried
monitor.html	Session/Client ID display	Web	Implemented	Wired	Static fields (demo-session-123, web-demo)	Not-inventoried
monitor.html	Overall progress bar + percent	Web	Implemented	Wired	Driven by WS translation_progress/step_completed events	Not-inventoried
monitor.html	Current step / step message panel	Web	Implemented	Wired	Shows current step + message from WS	Not-inventoried
monitor.html	Duration timer	Web	Implemented	Wired	Client-side elapsed clock	Not-inventoried
monitor.html	Status badge	Web	Implemented	Wired	Pending/Running/Completed/Failed	Not-inventoried
monitor.html	Event log	Web	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					Scrolling list of received WS events	Not-inver
monitor.html	Control Panel: Simulate Progress	Web	Implemented	Wired	Locally fakes & echoes progress events over WS (demo only)	Not-inver
monitor.html	Clear Log / Disconnect / Reconnect WS	Web	Implemented	Wired	Local log clear + WS connect/disconnect to ws://localhost:8090/ws	Not-inver
enhanced-monitor.html	Stats header (Active/Completed/Errors)	Web	Implemented	Wired	Aggregated counters from WS-tracked sessions	Not-inver
enhanced-monitor.html	Control panel: Connect/Disconnect/Clear	Web	Implemented	Wired	WS to ws://localhost:8090/ws?client_id=dashboard	Not-inver
enhanced-monitor.html	Monitor specific session	Web	Implemented	Wired	Input + button to filter UI to a given session_id	Not-inver
enhanced-monitor.html	Worker Information panel	Web	Implemented	Wired	Host:port, model, capacity from WS worker_info	Not-inver
enhanced-monitor.html	Current Session Progress panel	Web	Implemented	Wired	Progress bar, step, current/total items, duration, status	Not-inver
enhanced-monitor.html	Live progress chart (Chart.js)	Web	Implemented	Wired	Progress % over time, last 20 points	Not-inver
enhanced-monitor.html	Event log	Web	Implemented	Wired	Color-coded WS event stream	Not-inver
enhanced-monitor.html	Session History table	Web	Implemented	Wired	Last 10 sessions: id, worker, status, duration, progress, start	Not-inver
dashboard.html	Stats cards (Total/Completed/Running/Failed)	Web	Implemented	Wired	Polls GET /api/v1/translations every 5s	Not-inver
dashboard.html	Active Sessions table	Web	Implemented	Wired	Lists sessions w/ provider, status, progress, duration	Not-inver

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
dashboard.html	Session Details view	Web	Implemented	Wired	Progress bar, steps, generated files via GET /api/v1/translations/:id	Not-inver
dashboard.html	Live progress via WebSocket	Web	Implemented	Wired	Connects ws://localhost:8080/ws?session_id=...; progress_update/step_*	Not-inver
dashboard.html	New Translation modal	Web	Implemented	Wired	File input, provider selector, source/target lang, API-key/SSH/llama fields (UI only)	Not-inver
dashboard.html	Start Translation (submit)	Web	Implemented	Wired	POST /api/v1/translations with provider config — triggers backend translation	Not-inver
dashboard.html	Cancel session	Web	Implemented	Wired	DELETE /api/v1/translations/:id	Not-inver

Web Dashboard backend handler — pkg/api/dashboard.go

Backend that serves the dashboard page (/ , /dashboard, /monitor) and the translations endpoints. Was previously UNROUTED (returned 404); now wired so the dashboard genuinely translates for end users. Source: pkg/api/dashboard.go, commit f6ba5cc.

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
api/ dashboard.go	Dashboard page + translations endpoints	Web	Implemented	Wired (was Unwired/404 pre-fix)	Serves the dashboard HTML at / , /dashboard, /monitor and the GET/POST /api/v1/translations path; Start-Translation now runs a real translation through the wired UI path	Covered + ? RED→GR guard dashboard (default GR asserts page real translation RED_MODE= reproduces fix 404) — go test - pkg/api G this session

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
						(no longer a 404)

REST API — pkg/api/server.go (standalone Server, gin)

Component	Feature	Category	Implementation	Wiring	Real-use
api/server.go	GET /health	API	Implemented	Wired	Status + translator name
api/server.go	POST /api/translate	API	Implemented	Wired	translator.Translate(...)
api/server.go	GET /api/languages	API	Implemented	Wired	Static language list
api/server.go	GET /api/stats	API	Implemented	Wired	Returns hardcoded zeros (not live metrics)
api/server.go	POST /api/upload	API	Implemented	Wired	Save uploaded file, detect format
api/server.go	POST /api/batch	API	Partial	Wired	Returns batch_id queued only; performs NO translation

REST API — pkg/api/handler.go (primary Handler, gin)

Component	Feature	Category	Implementation	Wiring	Real-use
api/handler.go	GET /health, /, /api/v1/version, /api/v1/providers, /api/v1/stats	API	Implemented	Wired	Static/info cache stats
api/handler.go	GET /ws	API	Implemented	Wired	WebSockets to wsHub
api/handler.go	POST /api/v1/translate	API	Implemented	Wired	Distributed Translation

Component	Feature	Category	Implementation	Wiring	Real-use
					target_url was hardcoded
api/handler.go	POST /api/v1/translate/fb2	API	Implemented	Wired	Parse ebooks
api/handler.go	POST /api/v1/translate/batch	API	Implemented	Wired	Loops text
api/handler.go	POST /api/v1/convert/script	API	Implemented	Wired	Cyrillic[?] → Latin
api/handler.go	GET /api/v1/status/:session_id	API	Implemented	Wired	Returns health of a live session
api/handler.go	GET /api/v1/languages	API	Implemented	Wired	Static list of supported language.C
api/handler.go	POST /api/v1/translate/validate	API	Implemented	Wired	Validates n-grams (min length); not
api/handler.go	POST /api/v1/preparation/analyze	API	Stub	Wired	File stat/compute analysis, not
api/handler.go	GET /api/v1/preparation/result/:session_id	API	Stub	Wired	Returns metadata
api/handler.go	POST /api/v1/translate/ebook	API	Stub	Wired	Emits start of work; does NOT
api/handler.go	POST /api/v1/translate/cancel/:session_id	API	Implemented	Wired	Emits cancellation; cancelled
api/handler.go	GET /api/v1/distributed/status	API	Implemented	Wired	dm.GetStatus
api/handler.go	POST /api/v1/distributed/workers/discover	API	Implemented	Wired	SSH work
api/handler.go	POST /api/v1/distributed/workers/:worker_id/pair	API	Implemented	Wired	Pair worker
api/handler.go	DELETE /api/v1/distributed/workers/:worker_id/pair	API	Implemented	Wired	Unpair worker
api/handler.go	POST /api/v1/distributed/translate	API	Implemented	Wired	dm.Translate
api/handler.go	POST /api/v1/update/{upload,apply,rollback}	API	Implemented	Wired	Worker self-update; install/rollback
api/handler.go	GET /api/v1/monitoring/version/{metrics,alerts,health,dashboard}	API	Implemented	Wired	Version-synchronisation
api/handler.go	POST /api/v1/monitoring/version/drift-check	API	Implemented	Wired	Trigger drift-check
api/handler.go	GET /api/v1/monitoring/version/alerts/history	API	Implemented	Wired	Alert history
api/handler.go		API	Implemented	Wired	Ack alert

Component	Feature	Category	Implementation	Wiring	Real-use
	POST /api/v1/monitoring/ version/alerts/:alert_id/ acknowledge				
api/ handler.go	POST /api/v1/monitoring/ version/alerts/channels/ {email,webhook,slack}	API	Implemented	Wired	Add alert
api/ handler.go	GET /api/v1/monitoring/version/ dashboard.html	Web	Implemented	Wired	Serve emb
api/ handler.go	POST /api/v1/auth/login, /auth/ token	API	Implemented	Wired	JWT auth
api/ handler.go	GET /api/v1/profile	API	Implemented	Wired	Return cla middleware

REST API — pkg/api/batch_handlers.go (under /api/v1)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
api/ batch_handlers.go	POST / api/v1/ translate/ string	API	Implemented	Wired	NewLLMTranslator + Translate	Not- invent
api/ batch_handlers.go	POST / api/v1/ translate/ directory	API	Implemented	Wired	Batch processor translates each file	Not- invent

REST API — pkg/api/verifier_handlers.go (LLMsVerifier, under /api/v1)

Component	Feature	Category	Implementation	Wiring	Real-use
					List verified models (LLMsVerifier API call). GAP
api/ verifier_handlers.go	GET /api/ v1/verified- models	API	Implemented	Wired (only when LLMsVerifier enabled)	(§11.4.6): LLMsVerifier is DISABLED in the current config, so the route is not mounted and returns HTTP 404. Not a

Component	Feature	Category	Implementation	Wiring	Real-use
					defect — feature-disabled-by-config; needs a verifier-enabled config to confirm
api/verifier_handlers.go	GET /api/v1/verified-models/:id	API	Implemented	Wired	Get one verified model
api/verifier_handlers.go	GET /api/v1/verification-status	API	Implemented	Wired	Ping LLMsVerifier
api/verifier_handlers.go	POST /api/v1/verification/refresh	API	Implemented	Wired	Refresh scores
api/verifier_handlers.go	GET /api/v1/providers/verified	API	Implemented	Wired	Aggregated verified providers
api/verifier_handlers.go	POST /api/v1/translate-with-verification	API	Stub	Wired	Selects/verifies a model, returns metadata + text preview; does NOT translate

REST API — cmd/api-server/main.go (gRPC-proxying REST, :8080)

Component	Feature	Category	Implementation	Wiring	Real-use
api-server	POST /api/v1/translations	API	Implemented	Wired	Proxies grpcClient.StartTran
api-server	GET /api/v1/translations/:session_id	API	Implemented	Wired	Get status (proxy)
api-server	GET /api/v1/translations	API	Implemented	Wired	List (proxy)
api-server	DELETE /api/v1/translations/:session_id	API	Implemented	Wired	Cancel (proxy)
api-server	GET /api/v1/translations/:session_id/stream	API	Implemented	Wired	WS proxy of gRPC StreamTranslationP

Component	Feature	Category	Implementation	Wiring	Real-use
api-server	GET /api/v1/providers	API	Implemented	Wired	Get providers (prox
api-server	GET /api/v1/health	API	Implemented	Wired	Derived from gRPC state
api-server	GET /api/v1/metrics	API	Implemented	Wired	Returns static/zero m (not live)
api-server	GET /ws	Web	Implemented	Wired	WS upgrade, register wsHub (requires session_id)
api-server	GET /static/*	Web	Implemented	Wired	Static file server (./static)
api-server	GET /	Web	Implemented	Wired	Render dashboard.h template

WebSocket — hub & fan-out

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valid
websocket/hub.go	/ws endpoint (package mux)	Infra	Implemented	Wired	WS upgrade; reads client_id + session_id	Not-inventoried	Source confi
websocket/hub.go	Event fan-out	Infra	Implemented	Wired	Subscribes to ALL bus events; marshals events.Event JSON; per-session filter; Broadcast()	Not-inventoried	Source confi

gRPC — pkg/grpc/server.go (service TranslationService)

Component	Feature	Category	Implementation	Wiring	Real-use
grpc	StartTranslation	gRPC	Implemented	Wired	Reserves session spawns runTrans → translator.Transla
grpc	GetTranslationStatus	gRPC	Implemented	Wired	

Component	Feature	Category	Implementation	Wiring	Real-use
					Snapshot session translator.GetSta
grpc	ListTranslations	gRPC	Implemented	Wired	Iterate sessions
grpc	CancelTranslation	gRPC	Implemented	Wired	Cancel ctx + translator.Cancel
grpc	StreamTranslationProgress	gRPC	Implemented	Wired	Stream session progress events
grpc	GetProviders	gRPC	Implemented	Wired	Static ProviderRegistry (openai, anthropi ssh)
grpc	SubscribeEvents	gRPC	Implemented	Wired	Stream filtered SystemEvents fr bus

CORS / Middleware / Auth (Infra)

Component	Feature	Category	Implementation	Wiring	Real-use
cmd/server	corsMiddleware	Infra	Implemented	Wired	Driven by cfg.Security.CORSOrig reflects allowlisted Or + Allow-Credentials; wildcard * no-credentials OPTIONS→204
cmd/server	rateLimitMiddleware	Infra	Implemented	Wired	Per-ClientIP via security.RateLimiter; 4 on exceed
pkg/api/handler.go	authMiddleware	Infra	Implemented	Wired	Bearer JWT validation applied to protected subgroup, gated by Security.EnableAuth
pkg/api/server.go	authMiddleware (standalone)	Infra	Implemented	Wired	X-API-Key header == config.Security.APIKe Security.RequireAuth
pkg/grpc/server.go	message-size limits	Infra	Implemented	Wired	4 MB recv/send msg-s limits; no interceptors/ auth/rate-limit register

4. Owned Submodules

challenges/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
challenges	Challenge core engine	Submodule	Implemented	Wired	Challenge interface, Config, Result, BaseChallenge, ProgressReporter (liveness)	Not-inver
challenges	Registry + dependency ordering	Submodule	Implemented	Wired	Challenge registration, Kahn topological sort by deps	Not-inver
challenges	Runner / execution engine	Submodule	Implemented	Wired	Sequential/parallel/ pipeline execution, liveness/stall monitoring	Not-inver
challenges	Assertion engine	Submodule	Implemented	Wired	16 built-in evaluators (not_empty, contains, min_count, max_latency...) + custom	Not-inver
challenges	Report generation	Submodule	Implemented	Wired	Markdown / JSON / HTML reporters	Not-inver
challenges	Live monitoring	Submodule	Implemented	Wired	WebSocket dashboard, progress liveness	Not-inver
challenges	Metrics	Submodule	Implemented	Wired	Prometheus-compatible challenge metrics	Not-inver
challenges	HTTP client	Submodule	Implemented	Wired	Generic REST client with JWT auth, functional options	Not-inver
challenges	Challenge bank loader	Submodule	Implemented	Wired	Load challenge definitions from JSON/YAML	Not-inver
challenges	Plugin system	Submodule	Implemented	Wired	Custom challenge types + assertion plugins	Not-inver
challenges	Infra bridge	Submodule	Implemented	Wired	Bridge to digital.vasic.containers for test infra	Not-inver
challenges	User-flow automation	Submodule	Implemented	Wired	8 adapter interfaces / 21 impls (Playwright, ADB, Tauri, HTTP, gRPC, WS, Gradle, Cargo, npm), 19	Not-inver

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					templates, 12 evaluators	
challenges	userflow-runner CLI	Submodule	Implemented	Wired	CLI runner for user-flow challenges	Not-inventoried
challenges	panoptic / i18n / env / logging	Submodule	Implemented	Wired	Observation, i18n, env-var redaction, structured logging	Not-inventoried
challenges	anti_bluff lib	Submodule	Implemented	Wired	Shell anti-bluff helper library	Not-inventoried

containers/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
containers	boot manager	Submodule	Implemented	Wired	On-demand infra boot/start orchestration	Not-inventoried
containers	compose	Submodule	Implemented	Wired	docker/podman compose file orchestration	Not-inventoried
containers	health checking	Submodule	Implemented	Wired	TCP/HTTP health-check primitives	Not-inventoried
containers	runtime auto-detect	Submodule	Implemented	Wired	Docker/ Podman/ Kubernetes runtime detection	Not-inventoried
containers	endpoint	Submodule	Implemented	Wired	ServiceEndpoint builder	Not-inventoried
containers	lifecycle	Submodule	Implemented	Wired	Container lifecycle management	Not-inventoried
containers	service discovery / registry	Submodule	Implemented	Wired	Service discovery + registry	Not-inventoried
containers	orchestrator / scheduler	Submodule	Implemented	Wired	Orchestration + scheduling	Not-inventoried
containers	emulator suite	Submodule	Implemented	Wired	Android/device emulator boot/ matrix/cleanup/ canary	Not-inventoried
containers	genymotion	Submodule	Implemented	Wired	Genymotion device support	Not-inventoried
containers	vm-matrix	Submodule	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					VM matrix orchestration	Not-inventoried
containers	distributed build/test	Submodule	Implemented	Wired	Distributed build + test commands	Not-inventoried
containers	crossbuild	Submodule	Implemented	Wired	Cross-platform build	Not-inventoried
containers	ctop	Submodule	Implemented	Wired	Container top / monitoring TUI	Not-inventoried
containers	network / volume	Submodule	Implemented	Wired	Network + volume primitives	Not-inventoried
containers	supporting primitives	Submodule	Implemented	Wired	cache/metrics/monitor/event/policy/remote/lazyservice/brokertest/envconfig	Not-inventoried

doc_processor/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
doc_processor	Document loader	Submodule	Implemented	Wired	Loads Markdown / YAML / HTML docs (10 MB cap)	Not-inventoried
doc_processor	Feature-map builder	Submodule	Implemented	Wired	Builds structured FeatureMap; heuristic extraction + LLM Enrich	Not-inventoried
doc_processor	LLMAgent interface	Submodule	Implemented	Wired	Summarize/ExtractFeatures/ClassifyFeature/InferScreens/GenerateTestSteps	Not-inventoried
doc_processor	Coverage tracker	Submodule	Implemented	Wired	Verification coverage tracking by platform/category	Not-inventoried
doc_processor	DocGraph	Submodule	Implemented	Wired	Doc node/edge graph	Not-inventoried
doc_processor	Config / i18n	Submodule	Implemented	Wired	Configuration + i18n	Not-inventoried
doc_processor		Submodule	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
	docprocessor CLI				CLI running feature extraction against a docs dir	Not- inven
doc_processor	archdoc (internal)	Submodule	Implemented	Wired	Architecture-doc helper (LLM-use UNCONFIRMED)	Not- inven

docs_chain/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
docs_chain	sync command	Submodule	Implemented	Wired	Propagate changes through chain (regenerate + re-export atomically)	Not- inventori
docs_chain	verify command	Submodule	Implemented	Wired	Deterministic CI/ pre-build gate (chain in sync)	Not- inventori
docs_chain	doctor command	Submodule	Implemented	Wired	Diagnostics	Not- inventori
docs_chain	graph command	Submodule	Implemented	Wired	Show DAG / topo order	Not- inventori
docs_chain	watch daemon	Submodule	Implemented	Wired	fsnotify-driven continuous sync	Not- inventori
docs_chain	exec: transforms	Submodule	Implemented	Wired	Shell-out transforms (pandoc md→html/docx, weasyprint html→pdf)	Not- inventori
docs_chain	content- hash engine	Submodule	Implemented	Wired	Content-hash incremental recompute; roster/ corpus fingerprint (§11.4.86)	Not- inventori
docs_chain	DAG + recompute	Submodule	Implemented	Wired	Node/Edge/ Graph, cycle detection, Kahn topo, early-cutoff, sync-edge conflict	Not- inventori
docs_chain	adapters	Submodule	Implemented	Wired	FileAdapter, SQLiteAdapter (pure-Go modernc),	Not- inventori

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
docs_chain	orchestrator	Submodule	Implemented	Wired	DERIVED pandoc/ weasyprint (ToolAbsentError) Atomic run: staged writes, SQLite-txn commit + rollback	Not-inventori
docs_chain	DB builtins	Submodule	Implemented	Wired	Bidirectional md- to-sqlite / sqlite- to-md (row-level drift); colorize- html	Not-inventori
docs_chain	config + state	Submodule	Implemented	Wired	YAML multi- context loader; state.json	Not-inventori

llm_orchestrator/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
llm_orchestrator	Agent interface + AgentPool	Submodule	Implemented	Wired	Thread-safe pool, capability matching, acquire/ release, health	Not-invent
llm_orchestrator	CLI agent adapters	Submodule	Implemented	Wired	OpenCode, ClaudeCode, Gemini, Junie, QwenCode adapters (BaseAdapter)	Not-invent
llm_orchestrator	Transport / protocol	Submodule	Implemented	Wired	PipeTransport (JSON-line stdin/stdout) + FileTransport, protocol types	Not-invent
llm_orchestrator	Response parser	Submodule	Implemented	Wired	Parses agent JSON-line responses	Not-invent
llm_orchestrator	HealthMonitor + CircuitBreaker	Submodule	Implemented	Wired	Health tracking + fault tolerance over agents	Not-invent

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
llm_orchestrator	Config / i18n	Submodule	Implemented	Wired	Pool/agent config; i18n	Not-invent
llm_orchestrator	orchestrator CLI	Submodule	Implemented	Wired	CLI entry for orchestration	Not-invent

llm_provider/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
llm_provider	LLMProvider interface	Submodule	Implemented	Wired	Complete/CompleteStream/HealthCheck/GetCapabilities/ValidateConfig	Not-inver
llm_provider	Circuit breaker	Submodule	Implemented	Wired	closed/open/half-open fault tolerance	Not-inver
llm_provider	Health monitor	Submodule	Implemented	Wired	healthy/degraded/unhealthy/unknown	Not-inver
llm_provider	Retry logic	Submodule	Implemented	Wired	Exponential backoff + jitter, HTTP-status detection	Not-inver
llm_provider	Model discovery	Submodule	Implemented	Wired	Provider/model discovery (HTTP/endpoint-based)	Not-inver
llm_provider	API-key management	Submodule	Implemented	Wired	API key handling	Not-inver
llm_provider	HTTP client + models + i18n	Submodule	Implemented	Wired	Shared HTTP client, request/response models, i18n	Not-inver
llm_provider	Provider implementations (43)	Submodule	Implemented	Wired	ai21, anthropic, cerebras, chutes, claude, cloudflare, codestral, cohere, deepseek, fireworks, gemini, generic, githubmodels, groq, huggingface,	Not-inver

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					hyperbolic, junie, kilo, kimi, mistral, modal, nia, nlpcloud, novita, nvidia, ollama, openai, openrouter, perplexity, publicai, qwen, replicate, sambanova, sarvam, siliconflow, together, upstage, venice, vulavula, xai, zai, zen, zhipu	

llms_verifier/

Component	Feature	Category	Implementation	Wiring	Real-use	Te
llms_verifier	Model verification	Submodule	Implemented	Wired	“Do you see my code?” + existence/ responsiveness/ latency/streaming/ function-calling/ vision/embeddings tests	No inv
llms_verifier	Provider adapters	Submodule	Implemented	Wired	OpenAI, Anthropic, Cohere, Groq, Together, Mistral, xAI, Replicate, DeepSeek, Cerebras, Cloudflare, SiliconFlow (12+)	No inv
llms_verifier	Scoring engine	Submodule	Implemented	Wired	Quality scoring system (LLM-use UNCONFIRMED)	No inv
llms_verifier	Capabilities detection	Submodule	Implemented	Wired	Per-model capability discovery	No inv
llms_verifier	Failover / health monitoring	Submodule	Implemented	Wired	Real-time monitoring +	No inv

Component	Feature	Category	Implementation	Wiring	Real-use	Te
					intelligent failover (probes call LLM)	
llms_verifier	AI analytics / recommendations	Submodule	Implemented	Wired	AI-powered insights, trend analysis, model recommendations	No inv
llms_verifier	RAG / context mgmt	Submodule	Implemented	Wired	Vector DB semantic search, LLM-powered summarization	No inv
llms_verifier	LLMOps / self-improve / benchmark	Submodule	Implemented	Wired	LLM operations, self-improvement loop, benchmarking	No inv
llms_verifier	Messaging	Submodule	Implemented	Wired	Notification/messaging integration (Splunk/DataDog/ etc.)	No inv
llms_verifier	suffix / verified export	Submodule	Implemented	Wired	Mandatory branding suffix + verified-only config export	No inv
llms_verifier	Auth / security / API / API-keys	Submodule	Implemented	Wired	LDAP/SSO/ SAML/OIDC, SQLCipher, REST API, key mgmt	No inv
llms_verifier	Multi-platform clients	Submodule	Implemented	Wired	CLI, TUI, Web, Desktop, Mobile (UI; LLM-backed)	No inv
llms_verifier	helixqa / cliagents / crush / opencode	Submodule	Implemented	Wired	Integration packages (LLM-use UNCONFIRMED)	No inv
llms_verifier	scheduler / partners / bigdata / multimodal / performance	Submodule	Implemented	Wired	Scheduling, partners, big-data, multimodal, perf (multimodal likely LLM)	No inv
llms_verifier	main CLI + ACP	Submodule	Implemented	Wired	Main CLI; ACP (Agent Client Protocol)	No inv

security/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
security	Content guardrails	Submodule	Implemented	Wired	Rule pipeline (MaxLength, ForbiddenPatterns, RequireFormat) w/ severities	Not-invento
security	PII detection + redaction	Submodule	Implemented	Wired	Email/Phone/SSN/CreditCard(Luhn)/IPv4 detection; Mask/Hash/Remove	Not-invento
security	Content filtering	Submodule	Implemented	Wired	Composable chain: Length/Pattern/Keyword filters	Not-invento
security	Policy enforcement	Submodule	Implemented	Wired	Named policies, 8 condition operators, Allow/Deny/Audit + audit trail	Not-invento
security	Vulnerability scanner	Submodule	Implemented	Wired	Findings/reports, severity filtering + merging	Not-invento
security	HTTP security headers	Submodule	Implemented	Wired	Middleware: CSP/HSTS/X-Frame-Options + 7 standard headers	Not-invento
security	Secure storage	Submodule	Implemented	Wired	AES-256-GCM encrypted file KV store, Argon2id KDF	Not-invento
security	Privilege-escalation scanner	Submodule	Implemented	Wired	Probes /proc/self/status, capability bitmasks, namespace cgroups	Not-invento
security	SSRF denial	Submodule	Implemented	Wired	Denies SSRF on caller's outbound HTTP client	Not-invento
security	E2EE	Submodule	Implemented	Wired	End-to-end encryption	Not-invento
security	Attestation / GPU attestation	Submodule	Implemented	Wired	Attestation primitives	Not-invento
security	i18n	Submodule	Implemented	Wired	i18n	Not-invento

vision_engine/

Component	Feature	Category	Implementation	Wiring	Real-use
vision_engine	Analyzer interface	Submodule	Implemented	Wired	UI-element/text-region, issue detection, screen diff, key frames
vision_engine	LLM vision adapters	Submodule	Implemented	Wired	OpenAI GPT-4o, Anthr Claude, Gemini 2.0 Fla Qwen VL, Kimi, StepF Ollama + Fallback chain
vision_engine	NavigationGraph	Submodule	Implemented	Wired	Directed graph of screen transitions
vision_engine	Remote Ollama deployment	Submodule	Implemented	Wired	SSH auto-install, mode lifecycle; GPU/CPU/R detection; distributed ll RPC
vision_engine	OpenCV (stub + build-tag real)	Submodule	Partial	Wired	OpenCV-based CV — s default, real impl behind tag
vision_engine	Config / i18n	Submodule	Implemented	Wired	Configuration (HELIX_VISION_PRO etc.), i18n

Honest gaps & caveats (§11.4.6)

1. **Video coverage is 21 feature rows** (+ 1 real-artifact-confirmed). 21 features are confirmed by real, ffp_{ro}be- and content-verified recordings this session (see the Anti-bluff note for the full list); DOCX output is real-artifact-confirmed (produced .docx = Microsoft Word 2007+) with its video still PENDING. Every other runtime/user-visible feature is PENDING a recording. This is stated, not hidden. 1a. **DOCX output is Implemented (Rev 3, commit 87cd2be)** — pure-Go OOX_{ML} writer; a real DeepSeek run produced garden_es.docx = Microsoft Word 2007+. **PDF output is now Implemented (Rev 4, commit fb265e7)** — pkg/ebook/pdf_writer.go (Book→HTML5→weasyprint→Cyrillic-faithful PDF), wired into unified-translator (.pdf output, main.go:1160); pkg/ebook PDF tests green. Output formats are now EPUB/FB2/TXT/HTML/MD/DOCX/PDF. 1b. **-multipass ≠ -verify**. The unified-translator -verify flag runs the CLI’s own per-step output check; the separate -multipass flag (d53e085) invokes the pkg/verification multi-pass polisher engine — now wired and video-confirmed. 1c. **cmd/translator local path is a STUB** (“local translation not yet implemented”); its remote path is OPERATOR-BLOCKED (no SSH/llama.cpp worker host). 1d. **translate-ssh + ebook-translator are OPERATOR-BLOCKED** — both require a remote SSH worker host running llama.cpp; none available. Binaries build and run as far as possible (blocked-binaries demo

- recorded). 1e. **GET /api/v1/verified-models returns 404** — LLMsVerifier is disabled in the current config so the route is not mounted (feature-disabled-by-config, not a defect). 1f. **cmd/cli title path-leak FIXED** (d53e085) + **write-safety FIXED** (87cd2be, atomic write/no-clobber, guard cmd/cli/no_partial_output_test.go).
2. **Tests = Not-inventoried** for all rows: the raw inventory did not perform per-feature test attribution. The repo carries extensive `_test.go` + `test/` suites, but mapping each to a feature is future work — recorded as honest unknown, never guessed Covered/None.
 3. **4 STUB endpoints** return mock data and do NOT do the named work: `POST /api/v1/translate/ebook`, `POST /api/v1/preparation/analyze`, `GET /api/v1/preparation/result/:id`, `POST /api/v1/translate-with-verification`.
 4. **1 Partial endpoint:** standalone `pkg/api/server.go` `POST /api/batch` returns a queued `batch_id` only and performs no translation.
 5. **Static/hardcoded returns** (not stubs, but not live): `GET /api/stats` (zeros), `GET /api/v1/status/:id` (hardcoded completed), `GET /api/v1/metrics` (static), monitor-server `GET /api/v1/status` (static `monitoring_active`), several static language/provider lists. Marked Implemented with the static behaviour stated in Real-use.
 6. **vision_engine OpenCV** is Partial — stub by default, real CV only behind a build tag.
 7. **UNCONFIRMED LLM determinations** carried verbatim from the inventory for thin OpenAI-compatible providers, several `llms_verifier` packages, challenges/userflow evaluators, and `doc_processor/archdoc` — noted inline rather than asserted.